

Physics

1st Ans: 3rd Concave mirror

2nd Ans: 3rd 30cm, 1.5cm.

3rd Ans: 4th Both 2 and 3

4th Ans: Focus: The point in front of a spherical mirror where rays coming from infinity converge at. Its distance from the pole is half of the spherical mirror's radius.

4th Ans: Centre of curvature: It is the centre of the sphere, of which the spherical mirror is a part of.

5th Ans: $m = -\frac{1}{5}$ and $v = 18\text{cm}$

$$\Rightarrow \frac{v}{u} = -\frac{1}{5}$$

$$\Rightarrow \frac{18}{u} = -\frac{1}{5}$$

$$\Rightarrow -90\text{cm} = u$$

∴ Nature of the image

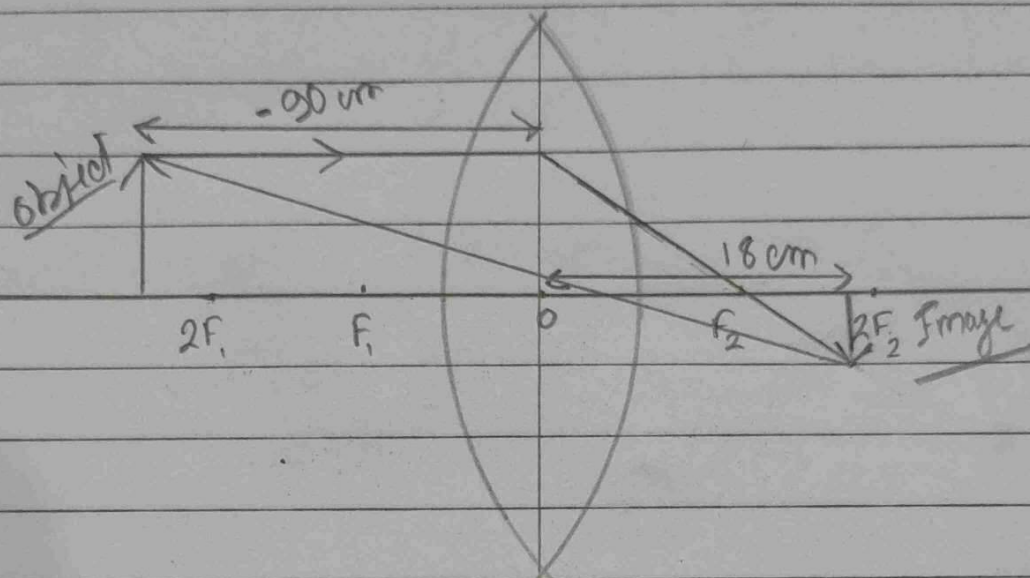
$$\text{Now, } \frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{18} - \left(-\frac{1}{90}\right) = \frac{1}{f}$$

$$\Rightarrow \frac{5+1}{90} = \frac{1}{f} \Rightarrow \frac{6}{90} = \frac{1}{f}$$

$\therefore f = 15 \text{ cm}.$

$\therefore$ Nature of lens is convex.



67 Ans 12:

$$u = -27 \text{ cm}$$

$$f = -18 \text{ cm}$$

$$\therefore \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{v} - \frac{1}{27} = -\frac{1}{18}$$

$$\Rightarrow \frac{1}{v} = -\frac{1}{18} + \frac{1}{27}$$

$$\Rightarrow \frac{1}{v} = \frac{-3+2}{54}$$

$$\Rightarrow \frac{1}{v} = -\frac{1}{54}$$

$$\Rightarrow v = -54 \text{ cm}$$

$\therefore$ ~~re~~ given, $h_o = 7 \text{ cm}$.

$$\text{Now, } m = \frac{(-v)}{u} = \frac{h_o}{h_i}$$

$$\Rightarrow \frac{-(-54)}{-27} = \frac{7}{h_i}$$

$$\Rightarrow \frac{-54}{-27} = \frac{7}{h_i}$$

$$\Rightarrow h_i = \frac{7}{2} \text{ cm}$$

Since, h_i is -ve

∴ Image is ~~virtual~~ real and inverted

& is two times larger than object.

Q7 ii) Ans:

$$u = 5 \text{ m}, f = 2 \times 5 = 10 \text{ m}$$

$$u = -20 \text{ m}$$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{v} - \frac{1}{20} = \frac{1}{10}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{10} + \frac{1}{20}$$

$$\Rightarrow \frac{1}{v} = \frac{2+1}{20}$$

$$\Rightarrow \frac{1}{v} = \frac{3}{20}$$

$$\Rightarrow v = \frac{20}{3} \text{ cm}$$

$$\text{Now } m = \frac{-v}{u}$$

$$= \frac{-\frac{20}{3}}{20}$$

$$= \frac{-20}{3} \times \frac{1}{-20} = \frac{1}{3}$$

$\therefore$ Image is virtual & erect and is $\frac{1}{3}$ times of the object.

6) iii) Ans The nature of the mirror used by the dentist is concave mirror. It is used, because it can form larger image of teeth and as a result the ~~dent~~ dentist can clearly see minute details inside the mouth of the patient.